Henry C. Wong

henrycwong.com
hcwong@uw.edu
309-750-0080
github.com/HenryCWong

RESERACH FOCUS Cyber-biosecurity, AI-driven genomic threat detection, and adversarial machine learning

EDUCATION

Ph.D in Computer Science — Sept 2024 - *Present*
University of Washington — Seattle, WA
Advisors: Tadayoshi Kohno & Jeff Nivala

B.S/M.S. in Computer Science — Aug 2020 - May 2022
Missouri University of Science and Technology — Rolla, MO
Thesis: *Man-in-the-Middle Attacks on MQTT based IoT networks*

RESEARCH

Henry Wong, Tadayoshi Kohno, Jeff Nivala. *DNAS-Bench: Deterministic Nucleic Acid Screener Benchmarking*. *In Submission* **Paper**

Henry Wong. *"Genetically Modifying Plants to Disrupt Autonomous Wardrones"* UW CSE Research Poster Showcase, October 2024. **Poster**

Henry Wong. *"Cost of Large Language Model Solutions"* 2023 ML/DL National Laboratory Workshop, July 2023. **Talk**

Henry Wong, Tie Luo. *"Man-in-the-Middle Attacks on MQTT-based IoT Using BERT based Adversarial Message Generation."* 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 3rd AIoT Workshop, August 2020. **Paper**

PROFESSIONAL EXPERIENCE

University of Washington — Seattle, WA
Ph.D Student — September 2024 - *Present*
- Created DNAS-Bench, a benchmarking framework for evaluating sequence screening tools against adversarial sequences
- Designed large-scale experimental pipelines to create genetically modified plants to perform adversarial computer vision

Sandia National Laboratories — Albuquerque, NM
Cyber Security/AI Researcher — June 2022 - April 2025
- Led red team evaluations of diverse systems to identify vulnerabilities
- Built prototype Agentic AI workflows at Sandia and other customers.
- Spearheaded bioinformatics and bio-related deep learning efforts
- Principal Investigator for multiple cybersecurity and AI research projects funded through internal and external laboratory programs

Washington University in St. Louis — St. Louis, MO
Computational Biology/Bioinformatics Intern — May 2021 - August 2021
- Built bioinformatics pipelines optimized specifically to the University's new compute cluster to process genomics data including Hi-C, RNA-seq, and ChIP-seq

AWARDS

UW College of Engineering Dean's Fellowship	2024
2nd Place at Missouri S&T Graduate Poster Competition	2022
26th in CTF Time United States Ratings (M57)	2017

SKILLS

Technologies: C/C++, Linux, Python, R, Julia, SQL, AWS
Libraries: Pandas, PyTorch, Sci-kit Learn, Tensorflow